

AKASHDIP MAHAPATRA

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PROFESSIONAL SUMMARY

Results-driven Site Reliability Engineer with a strong foundation in mechanical systems thinking, specializing in **infrastructure automation, CI/CD pipelines, and distributed systems**. Proven ability to **accelerate development velocity** and reduce operational toil through Python scripting, design highly reliable AWS cloud architectures, and optimize complex enterprise applications.

CORE COMPETENCIES

- **Security & Observability: (SRE):** Datadog, CloudWatch, Vulnerability Management (CVE patching), SLA/SLO tracking, incident management.
- **DevOps & CI/CD:** GitHub Actions (utilizing IAM Role ARNs via OIDC for passwordless authentication), Jenkins, Terraform (Infrastructure as Code), Advanced Git Version Control (resolving complex remote history mismatches, local vs. remote state reconciliation).
- **Programming & Scripting:** Python (Boto3, Data Structures & Algorithms), Bash, C, SQL, **AI-Assisted Development (Leveraging GenAI for script automation, code optimization, and RCA)**.
- **Cloud Infrastructure:** AWS (Lambda, ECR, ECS, S3, IAM, VPC, EventBridge, **Apache Kafka** (Amazon MSK), EC2, DynamoDB), & fundamentals of Azure (Azure DevOps) and Google Cloud Platform.
- **Container Orchestration & Cloud:** Docker, AWS ECS, ECR, AWS Serverless (Lambda, EventBridge), Apache Kafka (Amazon MSK).
- **Operating Systems & Networking:** Linux (Ubuntu, RHEL), TCP/IP, DNS, HTTP/HTTPS, Nginx, Wireshark, nmap.

PROFESSIONAL EXPERIENCE

1. **Tata Consultancy Services (TCS)** | Onsite *Data Engineer (DevOps & Infrastructure Focus under British Airways)* | Nov 2025 - Present
 - Design and maintain automated CI/CD deployment lifecycles utilizing GitHub Actions and Docker, enabling continuous delivery for multi-tiered AWS architectures processing **high-throughput, large-scale IoT data (200 GB to 500 GB daily)**.
 - Support, troubleshoot, and maintain high-throughput event-driven infrastructure (Amazon MSK/Kafka, EventBridge) ensuring strict environment consistency and system availability.
 - Apply SRE principles to container security by performing comprehensive Root Cause Analysis (RCA) on Docker images and Python dependencies, patching critical CVEs to improve system reliability, **enable self-healing system capabilities**, and minimize potential downtime (0 min for most of the Pipelines & for some maximum 30 min downtime).
 - Engineer and maintain a Python and Boto3-based **automation script** currently adopted by our 30-person DevOps team for universal task validation (including S3 data dumps, deployments, and infrastructure updates). Integrate with Datadog and CloudWatch to **monitor high-performance workflows and alert on low-latency applications**, successfully reducing manual operational toil by 90%, **significantly simplifying team operations, and accelerating development velocity** through continuous, feedback-driven iterations.
 - Successfully architected and onboarded the Pathfinder consumer to the Cabin Crew Ground activities, Details, and Schedules APIs, ensuring seamless data retrieval.
2. **Tata Consultancy Services (TCS)** | Remote *Network & DevOps Trainee (Apple Infrastructure Project)* | Oct 2025
 - Troubleshoot deep Linux OS and networking issues, analyzing **system calls (strace)**, **inodes**, and network traffic using TCP dump and Wireshark.
 - Deployed and configured Nginx reverse proxies and load balancers, successfully resolving HTTP 403/404 errors and DNS routing failures.
 - Hardened server security by implementing strict file permissions, writing custom Bash scripts for system automation, and configuring network firewalls (UFW, iptables, AWS Security Groups).

SYSTEMS ENGINEERING & AUTOMATION PROJECTS

1. Cloud Cost Optimization & Observability Investigation

- Spearheaded an extensive AWS cost anomaly investigation, utilizing CloudTrail and Datadog to identify operational inefficiencies across Lambda, SNS, and SQS services.
- Analyzed system behaviors and resource allocation to provide actionable insights for reducing cloud spend and optimizing infrastructure performance.

2. Lightweight Python Web Server (OSI Layer 4/7):

- Bypassed standard web frameworks to build a raw HTTP server from scratch using basic Python TCP Socket Programming, handling low-level GET/POST request parsing.

3. Greenfield Enterprise Cloud Architecture & Provisioning

- **Collaborate** with cross-functional development and DevOps teams to architect and provision a completely new cloud infrastructure following the decommissioning of the legacy Mercury 2 (M2) environment.
- Participate in daily system design Knowledge Transfer (KT) sessions to define scalable networking, security, and deployment standards for the greenfield environment.

4. Data Pipeline & Release Automation (GSE Events)

- Developed and deployed Python-based software solutions utilizing Boto3 to automate complex validation workflows, integrating APIs to eliminate manual processes and accelerate software release cycles.
- Orchestrated **high-throughput** data extraction workflows bridging legacy Oracle SQL databases with AWS S3 cloud storage, ensuring reliable, **performant data ingestion for downstream analytics and data processing teams**.

5. Network Security & Packet Forensics:

- Engineered a Python-based wireless penetration testing system to scan 802.11 traffic, capture handshakes, and test access point vulnerabilities.

PUBLICATIONS & APPLIED RESEARCH (Data Science & Machine Learning)

Modeling and Optimization of Surface Roughness of Electrodeposited Nickel Coating using Taguchi and Bonobo Optimizer | Jadavpur University Press (ISBN: 978-81-993635-8-8)

- Applied the Bonobo Optimizer (a meta-heuristic optimization algorithm) and Taguchi design of experiments to model, predict, and optimize complex physical processes.
- Demonstrated advanced capability in data-driven mathematical modeling, statistical analysis, and implementing optimization algorithms to solve multi-variable engineering challenges.

EDUCATION

Academy of Technology Engineering College | Hooghly, West Bengal Bachelor of Technology in Mechanical Engineering | Sep 2021 - 2025

- Applied mechanical systems-design principles and analytical problem-solving skills to complex cloud infrastructure and software architecture.
- **Mechanical Capabilities & Projects:** AutoCAD, SolidWorks (Advanced CAD, Sheet-Metal, Weldments), Ansys, MATLAB | **Key Focus:** Thermodynamics, Solid/Fluid Mechanics | **Applied Work:** 3D-printed working drone, fully functional V8 engine design, published academic research.

CERTIFICATIONS & QUALIFICATIONS

1. AWS Certified Cloud Practitioner (**Perfect Score: 1000/1000**)
2. Microsoft Certified: Azure Network Engineer Associate
3. Google Cloud Certified Professional
4. Oracle Cloud Infrastructure 2024 Certified Foundations Associate
5. GitHub Certifications: Advanced Security, Actions, and Administration
6. Postman Student Expert
7. NASA Open Science 101 Qualification.